

SciVer Router Models For Local VLMs

Public-review-safe summary of a three-candidate router over Qwen3-VL, Pixtral12B, and Kimi-VL-A3B using cognitive dimensions, chart TDA features, and claim NLP features.

Abstract

We study whether a lightweight router can choose among three local vision-language models for SciVer chart-claim verification. The router is trained on sanitized derived features and observed five-run local VLM accuracies, not raw claims, captions, images, prompts, or responses. We compare Ridge/ElasticNet regression, XGBoost, and PyTorch MLP regressors that predict each candidate VLM's accuracy and route to the highest predicted value. The best grouped-holdout point estimate is XGBoost with chart TDA features under mean-regret-first selection: 49.2% top-1 hit rate and 0.243 mean regret. Bootstrap intervals and overfitting curves show the result is promising but still uncertain on an 811-example dataset.

Rows

811

Best point-estimate top-1

49.2%

Best point-estimate regret

0.243

Held-out router rows

120

Read This First

Headline result: XGBoost with chart TDA has the best point estimate on the grouped held-out router split: 49.2% top-1 hit rate (40.5% - 58.4% bootstrap interval) and 0.243 mean regret (0.193 - 0.297 bootstrap interval). This headline is selected after comparing the public router grid, so the bootstrap interval is conditional on the selected run and does not adjust for model, feature-set, or selection-policy search. Treat it as internal model-development evidence, not an official untouched SciVer benchmark claim.

The public source labels `val` and `test` are original SciVer split-origin labels retained for reporting. Router evaluation uses the separate grouped `router_split` in each run's `splits.csv`.

What SciVer Evaluates

[SciVer: Evaluating Foundation Models for Multimodal Scientific Claim Verification](#)

introduces a benchmark for verifying scientific claims against multimodal evidence from papers, including figures and tables. The released [SciVer dataset](#) is licensed CC BY 4.0. This router study focuses on the chart-claim subset where each example can be represented by chart-derived difficulty features and claim-side semantic features.

Public reproducibility here means reproducing router training and evaluation from sanitized derived features. It does not mean rerunning private local VLM inference, exposing raw SciVer claims/captions, or publishing raw model outputs.

Key Definitions

- **Observed five-run accuracy:** for each chart-claim item and VLM, the fraction of five local runs that produced the correct SciVer label. Because each target uses only five repeats, per-item best-model labels and regret values are noisy, especially when VLMs tie.
- **Router target:** train one regression target per VLM accuracy, then select the VLM with the highest predicted accuracy.
- **Top-1 hit rate:** fraction of held-out examples where the routed VLM is one of the observed best VLMs, counting ties as hits.
- **Regret:** observed best accuracy minus the selected VLM's observed accuracy. Lower is better; zero means the router picked a best VLM.
- **Train-global-best baseline:** always pick the VLM with the highest mean accuracy on the router training split.
- **Grouped 10-fold CV:** split the training paper groups into 10 folds; for each candidate, train on 9 folds, predict the held-out fold, concatenate out-of-fold predictions, and rank candidates by the chosen CV policy. Every row from the same `paper_id` stays in one fold.
- **Grouped bootstrap interval:** after model selection, repeatedly resample held-out test paper groups to estimate uncertainty in the final test top-1 hit rate and mean regret. These intervals capture held-out group sampling uncertainty, not uncertainty from the five-repeat VLM accuracy estimates.

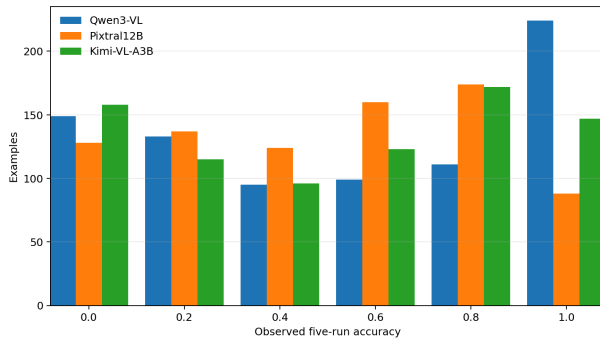
Data Generation And Public Dataset

We ran the three local VLMs on analytical/direct single-chart examples, used five repeats per example, and then sanitized the derived router dataset. The public dataset keeps IDs, paper groups, source split, three observed accuracy columns, best-model target columns, twelve cognitive dimensions, engineered scalar features, chart TDA features, and claim NLP features.

quantity	VLM	mean ob
public examples	Qwen3-VL	53.9%
validation-origin rows	Pixtral12B	49.3%
test-origin rows	Kimi-VL-A3B	51.8%
paper groups		
direct claims		
analytical claims		
public feature columns		

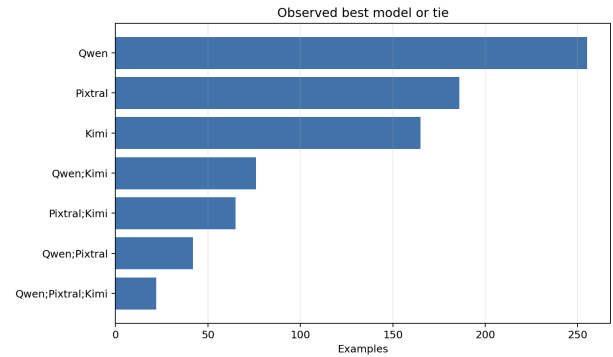
Best-model ties are common in this dataset: 606 examples have a single best VLM, 183 have two-way ties, and 22 have three-way ties. This is why top-1 hit rate is tie-aware rather than ordinary single-label accuracy.

Observed VLM Accuracy Distributions



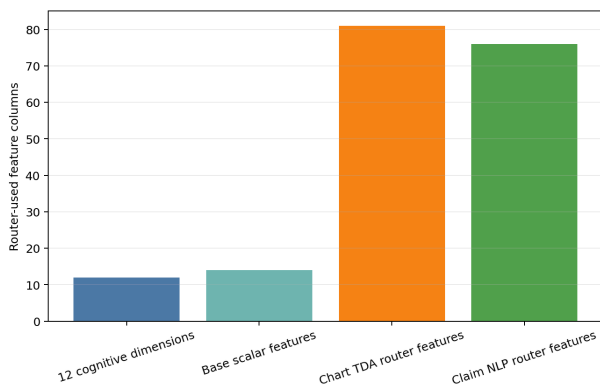
Because each item has five local repeats, observed accuracy can only take the six discrete values [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]. Each group of bars shows how many chart-claim items fall at that accuracy value for each VLM.

Observed Best VLM Or Tie



Counts show which VLM or tie-set has the highest observed five-run accuracy for an item. Best-model targets are recomputed only over Qwen3-VL, Pixtral12B, and Kimi-VL-A3B.

Feature Families In The Public Dataset



Counts show numeric feature columns used by the router after excluding TDA/NLP status metadata. Cognitive dimensions are human-coded annotations; chart TDA and claim NLP features are precomputed derived features.

Feature Construction

- **12 cognitive dimensions:** human-coded chart difficulty annotations such as visual load, arithmetic transformation, graphical scale, and knowledge needs. They are available in this public modeling table as precomputed features; a live deployment would need those annotations or an automatic proxy.
- **Base scalar features:** claim/caption length, numeric-token counts, percent counts, math-symbol counts, context length, and image dimensions.
- **Chart TDA:** persistent homology summaries from grayscale and edge filtrations, designed to capture visual structural complexity.
- **Claim NLP:** TF-IDF/SVD semantic coordinates, claim-caption alignment, reasoning cue counts, numeric overlap, and token-point-cloud TDA. The public NLP features were fit once on the full feature-extraction corpus before the router split, so they should be read as transductive exploratory features rather than a strict train-fold-only text pipeline.

feature family	public columns	meaning
12 cognitive dimensions	12	Human-coded chart difficulty annotations included as precomputed features; deployment would require these annotations or an automatic proxy.
Base scalar features	14	Simple size, token, numeric, and image-shape summaries.
Chart TDA	86	Topological summaries of grayscale and edge-map chart structure.
Claim NLP	81	TF-IDF/SVD semantics, cue counts, alignment, numeric overlap, and token-level TDA.

The public sidecar feature files can contain more rows than the 811-row modeling table because they are exported before complete-case filtering over the three public VLM accuracy columns. Router training uses the merged complete-case public dataset.

Acknowledgment: the 12 cognitive-difficulty dimensions were engineered by the [AI Faithfulness in Scientific Reasoning](#) project. This release attributes that contribution and does not claim to relicense upstream project code or documentation.

Router Workflow

1. Build the matched, sanitized three-VLM router dataset.
2. Construct base, chart-TDA, claim-NLP, and combined feature sets.
3. Split by `paper_id` into grouped train/validation/test sets.
4. Select hyperparameters by grouped 10-fold CV using either mean-regret-first or top-1-first ranking.
5. Refit the selected hyperparameters on train+validation and evaluate once on grouped test.
6. Train a final all-data model for future use after evaluation artifacts are written.

router split	rows	paper groups
train	568	434
validation	123	95
test	120	90
grouped CV	10 folds inside train	paper_id kept intact
bootstrap	1,000 held-out resamples	resampled with replacement
seed	215	

Candidate ranking is policy-dependent: `regret_first` sorts by lower grouped-CV mean regret, then higher top-1 hit rate, then lower complexity and stable candidate ID; `top1_first` swaps the first two criteria. The held-out test split is not used during candidate ranking inside any run, but the headline “best” run is chosen post-hoc after comparing the public grid of router families, feature sets, and selection policies.

The grouped CV implementation is deliberately stricter than row-random CV: it first restricts model selection to the router training split, then assigns unique `paper_id` groups to 10 folds. For each hyperparameter candidate, the script trains on nine paper-group folds, predicts the remaining fold, repeats this for all folds, and computes CV top-1

hit rate and mean regret from the concatenated out-of-fold predictions. This prevents related examples from the same paper from appearing on both sides of a CV fold.

Evaluation Results

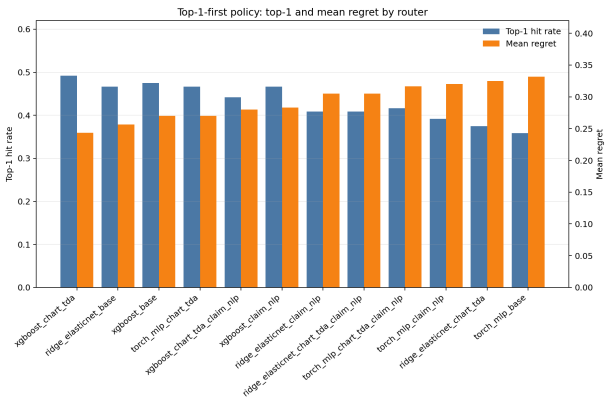
The strongest point-estimate result in the public artifact tree is **xgboost_chart_tda_regret_first**, with 49.2% held-out top-1 hit rate and 0.243 mean regret. The deterministic train-global-best baseline reaches 40.8% top-1 and 0.305 regret; the seeded random baseline reaches 38.3% top-1 and 0.295 regret. Baseline comparisons are shown as point estimates; baseline deltas are not separately uncertainty-tested. This keeps deterministic and stochastic baselines separate.

Regret-First Model Comparison



Bars show held-out top-1 hit rate and mean regret for all router families/features after hyperparameters are selected by lower grouped-CV mean regret first.

Top-1-First Model Comparison



Bars show the same held-out metrics after selecting candidates by higher grouped-CV top-1 hit rate first. Changing the selection priority does not displace XGBoost + chart TDA as the best point-estimate run.

Regret-First Metrics

run	router	features	selection	te to
xgboost_chart_tda_regret_first	xgboost	base_tda	regret_first	41
ridge_elasticnet_base_regret_first	ridge_elasticnet	base	regret_first	41

run	router	features	selection	test
xgboost_claim_nlp_regret_first	xgboost	base_nlp	regret_first	4%
xgboost_base_regret_first	xgboost	base	regret_first	4%
xgboost_chart_tda_claim_nlp_regret_first	xgboost	base_tda_nlp	regret_first	4%
ridge_elasticnet_claim_nlp_regret_first	ridge_elasticnet	base_nlp	regret_first	4%
torch_mlp_chart_tda_claim_nlp_regret_first	torch_mlp	base_tda_nlp	regret_first	4%
ridge_elasticnet_chart_tda_claim_nlp_regret_first	ridge_elasticnet	base_tda_nlp	regret_first	3%
torch_mlp_claim_nlp_regret_first	torch_mlp	base_nlp	regret_first	3%
ridge_elasticnet_chart_tda_regret_first	ridge_elasticnet	base_tda	regret_first	3%
torch_mlp_base_regret_first	torch_mlp	base	regret_first	3%
torch_mlp_chart_tda_regret_first	torch_mlp	base_tda	regret_first	3%

Top-1-First Metrics

run	router	features	selection	test top-1
xgboost_chart_tda_top1_first	xgboost	base_tda	top1_first	49.2%
ridge_elasticnet_base_top1_first	ridge_elasticnet	base	top1_first	46.7%
xgboost_base_top1_first	xgboost	base	top1_first	47.5%
torch_mlp_chart_tda_top1_first	torch_mlp	base_tda	top1_first	46.7%
xgboost_chart_tda_claim_nlp_top1_first	xgboost	base_tda_nlp	top1_first	44.2%
xgboost_claim_nlp_top1_first	xgboost	base_nlp	top1_first	46.7%
ridge_elasticnet_claim_nlp_top1_first	ridge_elasticnet	base_nlp	top1_first	40.8%
ridge_elasticnet_chart_tda_claim_nlp_top1_first	ridge_elasticnet	base_tda_nlp	top1_first	40.8%
torch_mlp_chart_tda_claim_nlp_top1_first	torch_mlp	base_tda_nlp	top1_first	41.7%
torch_mlp_claim_nlp_top1_first	torch_mlp	base_nlp	top1_first	39.2%

run	router	features	selection	test top-1
ridge_elasticnet_chart_tda_top1_first	ridge_elasticnet	base_tda	top1_first	37.5%
torch_mlp_base_top1_first	torch_mlp	base	top1_first	35.8%

Baselines For The Best Run's Split

baseline	type	test top-1	test mean regret	selected models
always_pixtral12b	deterministic	41.7%	0.302	pixtral12b: 120
always_qwen3vl	deterministic	40.8%	0.305	qwen3vl: 120
train_global_best	deterministic	40.8%	0.305	qwen3vl: 120
always_kimi_vl_a3b	deterministic	40.8%	0.315	kimi_vl_a3b: 120
random	stochastic	38.3%	0.295	kimi_vl_a3b: 38, pixtral12b: 38, qwen3vl: 44

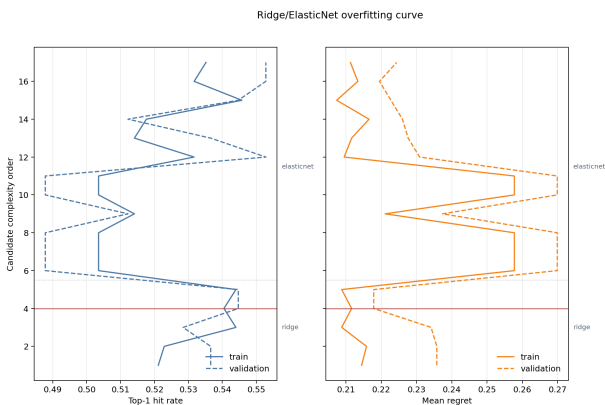
Selected-Model Counts

run	qwen3vl	pixtral12b	kimi_vl_a3b
regret_first/xgboost_chart_tda	56	33	31
regret_first/ridge_elasticnet_base	65	14	41
top1_first/torch_mlp_chart_tda	49	40	31

Overfitting Curves

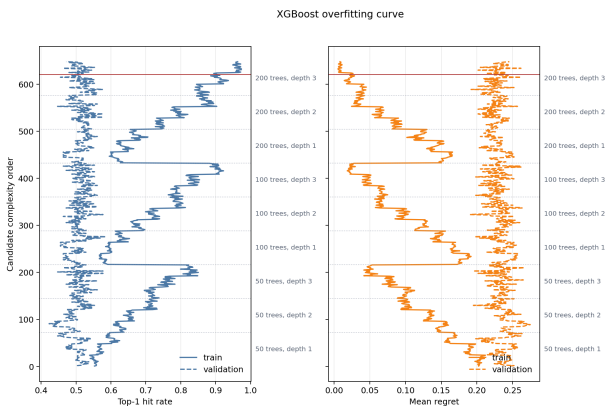
Overfitting curves compare training and validation metrics across candidate complexity. A widening train-validation gap, especially high training top-1 paired with lower validation top-1 or low training regret paired with higher validation regret, suggests the model is fitting patterns that do not generalize to held-out paper groups.

Ridge/ElasticNet Overfitting Curve



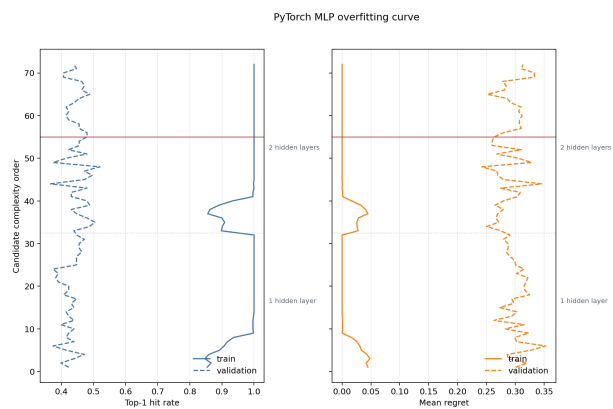
Curves show train versus grouped-validation top-1 and mean regret across regularization candidates for the best regret-first Ridge/ElasticNet run. The solid red horizontal line marks the selected candidate. Dotted horizontal separators mark Ridge versus ElasticNet candidate zones when both appear.

XGBoost Overfitting Curve



Curves show train versus grouped-validation metrics across the XGBoost grid. Dotted horizontal separators split tree-count/depth zones such as 50 trees depth 1, 50 trees depth 2, and so on; the right-side labels mark those zones. The solid red line marks the selected candidate.

PyTorch MLP Overfitting Curve



Curves show train versus grouped-validation metrics across the PyTorch MLP grid. Dotted horizontal separators split one-hidden-layer and two-hidden-layer candidates; the solid red line marks the selected candidate.

Selected-Candidate Overfitting Summary

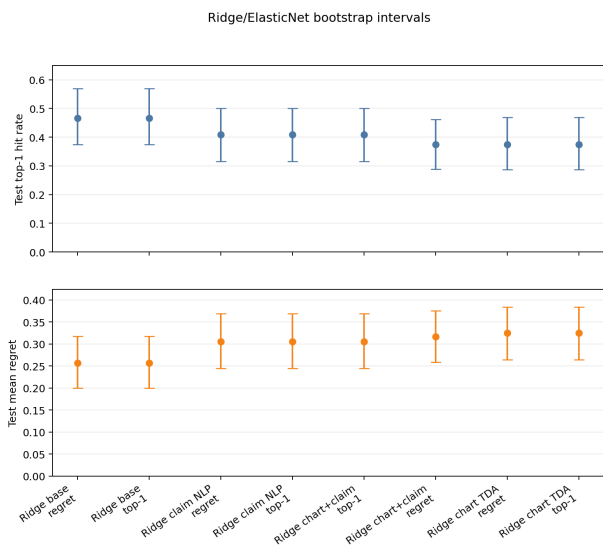
router family	selected run	selected candidate
Ridge/ElasticNet	ridge_elasticnet_base	ridge_alpha=0.1
XGBoost	xgboost_chart_tda	xgb_n=200_depth=3_lr=0.05_sub=1.0_col=1.0_l2=
PyTorch MLP	torch_mlp_chart_tda_claim_nlp	mlp_layers=32x16_act=tanh_alpha=0.01

The selected Ridge/ElasticNet and XGBoost candidates show smaller train-validation gaps than the selected PyTorch MLP candidate. This suggests limited-to-moderate overfitting risk for the linear and tree routers, while the MLP has the clearest overfitting signal: it can fit the training split much better than it validates on held-out paper groups. These curves diagnose the model-selection stage; final uncertainty is still judged on the separate grouped test split.

Bootstrap Uncertainty

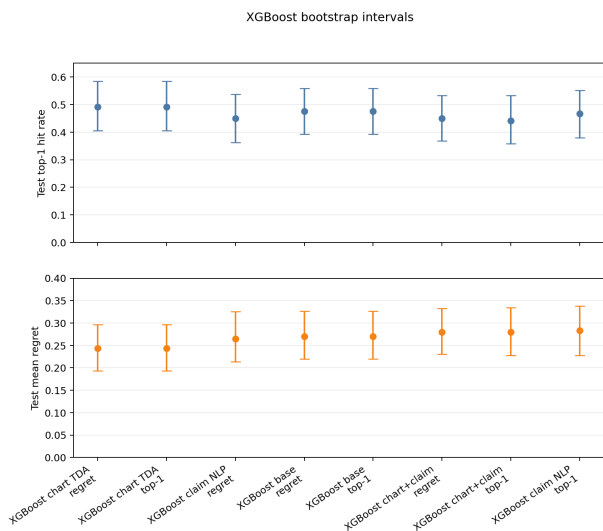
We resample only the held-out router test split because the bootstrap is estimating uncertainty for final reported performance after hyperparameters and model families have already been selected. Resampling training or validation rows would mix model-fitting/model-selection data into the uncertainty estimate. The implementation samples held-out `paper_id` groups with replacement 1,000 times, recomputes top-1 hit rate and mean regret for each resample, and reports percentile intervals. It does not refit the router and it does not account for five-repeat target noise or post-hoc winner selection across the public grid.

Ridge/ElasticNet Bootstrap Intervals



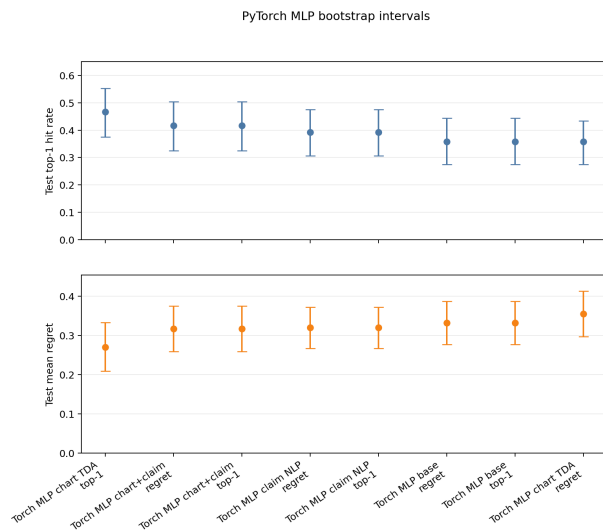
Each point is a held-out point estimate; whiskers are 95% grouped-bootstrap intervals from 1,000 resamples of test paper groups. The figure includes Ridge/ElasticNet runs across feature sets and both selection policies.

XGBoost Bootstrap Intervals



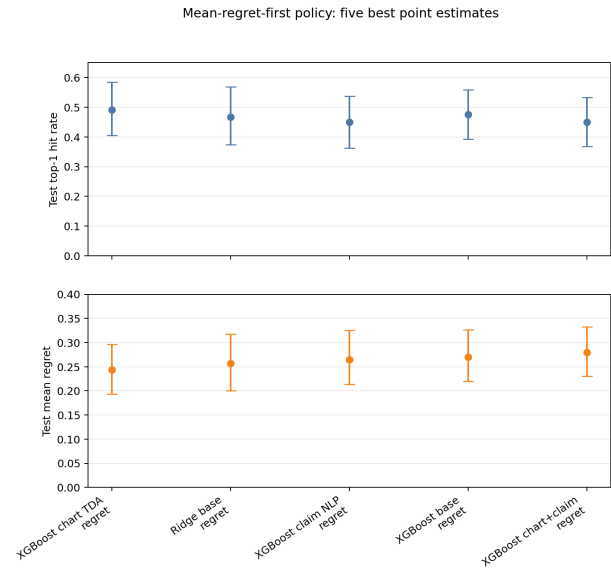
Each point is a held-out point estimate; whiskers are 95% grouped-bootstrap intervals. XGBoost + chart TDA has the best mean-regret point estimate, but intervals overlap with nearby candidates.

PyTorch MLP Bootstrap Intervals



Each point is a held-out point estimate; whiskers are 95% grouped-bootstrap intervals. MLP variants show substantial interval overlap, consistent with the small 120-row grouped holdout.

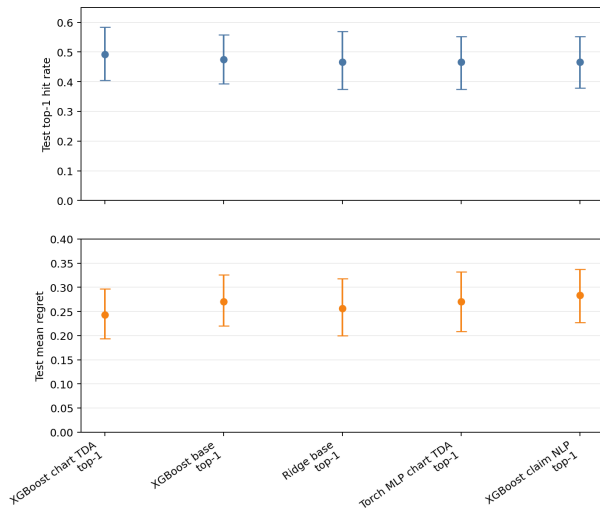
Mean-Regret-First Top Five



Only the five best point-estimate runs under the mean-regret-first policy are shown, sorted by lower held-out mean regret. Top panel shows tie-aware top-1; bottom panel shows mean regret.

Top-1-First Top Five

Top-1-first policy: five best point estimates



Only the five best point-estimate runs under the top-1-first policy are shown, sorted by higher held-out top-1 hit rate. Intervals are grouped-bootstrap intervals over held-out paper groups.

Supported Conclusions

- **Chart TDA helps most clearly for XGBoost as a point estimate:** XGBoost + chart TDA yielded the best grouped-holdout point estimate in this run, with mean regret 0.243 and top-1 hit rate 49.2%, but bootstrap intervals overlap with nearby candidates.
- **Claim NLP alone does not win in this run:** XGBoost + claim NLP reaches 45.0% top-1 and 0.265 regret, below chart TDA.
- **The router is behaviorally different from always choosing one VLM:** the best router routes held-out items across all three VLMs, while train-global-best always selects Qwen3-VL. This is a routing-diversity observation, not a claim that the performance difference is statistically separated.
- **Top-1-first selection is not automatically better:** the best top-1-first run ties the regret-first best point estimate, while several top-1-first NLP variants increase regret.
- **Uncertainty remains substantial:** bootstrap intervals are broad on 120 held-out examples and do not include five-repeat target noise or post-hoc grid-selection uncertainty, so conclusions should be presented as internal model-development point estimates.

Reproducibility

Primary public data artifact:

`Data/derived/model_router/public_router/router_dataset_public.csv` .

Public reviewers can reproduce router modeling from the bundled sanitized CSV without rebuilding private local VLM inference traces.

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
.venv/bin/python scripts/verify_public_router_release.py --root .
.venv/bin/python scripts/run_public_router_experiments.py
.venv/bin/python scripts/build_sciver_router_presentation_report.py
.venv/bin/python scripts/verify_public_router_release.py --root .
```

Use `--skip-existing` only for a quick check of bundled generated artifacts. For a true rerun, start from a fresh public release copy and omit it.

See `docs/reproduce_router_experiments.md` and `docs/public_data_statement.md` for the public/private boundary. Generated `.joblib` and `.pt` model files are included as artifacts, but loading serialized model files from untrusted sources can execute code; rerunning scripts is the recommended validation path. For exact-version reproduction, use `requirements-lock.txt` .

Generated from derived public router artifacts. No raw claims, captions, prompts, responses, images, SQLite annotations, or SciVer gold labels are rendered in this report.